CodeXplus Rebuild — A to Z Plan (Antigravity CLI)

Version 1 · 20 Oct 2026 · Fastify + PostgreSQL + custom Admin + Next.js (Webangles layout)

0. Is document ko kaise use karein

- Part 1 (sections 1–8): plan, architecture aur contracts. Part 2 (sections 9–12): API, admin, web spec aur saare prompts. Part 3 (sections 13–17): poora content, TODO list aur launch checklist.
- Prompts English mein hain taake `agy` mein seedha paste ho sakein. Jahan `[paste ...]` likha hai wahan pichle step ka output lagayein.
- Is document ko Markdown export karke repo mein `docs/PLAN.md` ke naam se rakh dein, taake `agy` khud parh sake.

1. Goal aur decisions

Goal: https://codexplus.site/ ka content Webangles Framer template (https://webangles.framer.website/) ke layout par laana, aur Sanity ki jagah apna Fastify backend + custom admin panel banana jis se poori website control ho.

Aapke final decisions

- Frontend layout: Webangles template.
- Backend: Fastify.
- Admin: custom panel jo poori website ko Sanity ki tarah control kare.
- Build tool: Antigravity CLI (`agy`).

Meri recommendations (chahein to change karein): pnpm monorepo, PostgreSQL + Drizzle ORM, Next.js 15 (public site), Vite + React (admin), zod shared schemas.

2. Architecture

```
codexplus/                  (pnpm workspace)
├─ apps/
│  ├─ api/       Fastify 5 + TypeScript (REST, auth, media, forms)
│  ├─ admin/     Vite + React admin panel (static build)
│  └─ web/       Next.js 15 public site (Webangles layout)
├─ packages/
│  └─ shared/    zod schemas + inferred types (single source of truth)
├─ content/seed/ migrated content (JSON) + downloaded assets
├─ design-ref/   tokens.json, sections.md, components.md, screenshots
├─ docs/         PLAN.md, api.md, contract-requests.md
├─ .agents/      skills/, mcp_config.json
├─ AGENTS.md
└─ PROGRESS.md
```

Domains: `www.codexplus.site` (web), `admin.codexplus.site` (admin static build), `api.codexplus.site` (Fastify, uploads bhi yahin se ya alag media subdomain se).

Layer	Choice	Kyun
API	Fastify 5 + fastify-type-provider-zod	Validation aur types ek hi schema se
DB	PostgreSQL + Drizzle ORM	Typed queries, migrations, JSONB blocks
Auth	argon2id, httpOnly cookies (access + refresh rotation), roles admin/editor	Token localStorage mein nahi
Media	@fastify/multipart + sharp	webp/avif variants + blurhash
Email	nodemailer (SMTP)	Contact notifications
Admin	Vite, React, shadcn/ui, TanStack Router + Query, react-hook-form, Tiptap, dnd-kit	Sanity Studio jaisa feel
Web	Next.js 15 App Router, Tailwind, Framer Motion, ISR + on-demand revalidate	Public site ko SEO ke liye server render chahiye
Tests	vitest, Playwright	API tests + visual/smoke tests
Deploy	Ubuntu, Nginx, PM2 (ya systemd), certbot, GitHub Actions over SSH	Aapka maujooda server workflow

Sabse important idea: `packages/shared` mein har content type aur har section block ka zod schema hoga. API validation, admin forms aur frontend types sab isi se banenge. Naya field ek jagah add hoga aur teeno jagah aa jayega.

Sanity jaisa control kaise milega

- Singletons: site settings, navigation, footer.
- Collections: projects, services, testimonials, posts, team, clients, tech stack, faqs.
- Pages + sections: home page ke sections ko reorder, on/off aur edit kar sakenge (JSONB blocks, har block ka apna schema).
- Draft / publish, preview token, revisions, media library, audit log.
- Publish par API signed webhook se Next.js ka cache revalidate karti hai.

3. Pehle fix karne wale issues

Yeh sab codexplus.site ko dekh kar mile hain. Inka faisla build shuru karne se pehle (ya Phase 5 tak) kar lein.

1. **Counters:** About section ke Projects Completed, Client Retention aur Global Awards ke numbers fetch mein 0 aaye (animated counters). Asli numbers chahiye. Sirf `12+ years` aur `40k+ creators shipped` nazar aaye.
2. **Contact placeholders:** intro lorem ipsum hai, website `www.reallygreatsite.com`, address `123 Anywhere St., San Francisco`, phone `+1 (415) 555-0132`. Yeh template ka placeholder lagta hai. Asli data footer mein hai: Karachi, `+92-3243315524`.
3. **Email mismatch:** `hello@codexplus.studio` (contact/menu) aur `info@codexplus.site` (CTA). Ek final karein.
4. **Purana badge:** `Currently booking Q3 2026 projects` ab stale hai (aaj 20 Oct 2026). Quarter update karein.
5. **Tech stack text:** Stack section mein Sanity CMS, Next API Routes aur Vercel likha hai. Naye stack (Fastify, custom CMS, Ubuntu/Nginx) ke hisaab se kya dikhana hai, yeh decide karein.
6. **Sanity CDN assets:** logo, og:image aur shayad project/article images `cdn.sanity.io` par hain. Sanity band karne se pehle sab download karke apni media library mein daalein (A6, P10).
7. **Template leftovers:** Webangles ke text mein Nubien, Reboot, SYNC, Mandro Design, Westhill Studio jaise purane naam hain. Inme se kuch bhi copy nahi karna, sirf layout lena hai.
8. **Claims aur testimonials:** client names, testimonials aur numbers (jaise `Conversion up 38%`) confirm karein ke asli hain aur publish ki ijazat hai.
9. **Social links:** footer ke links generic hain (`github.com`, `linkedin.com`, `twitter.com`, `instagram.com`, `dribbble.com`). Asli profile URLs chahiye.

10. **Subpages:** `/portfolio`, `/services`, `/team`, `/careers`, `/articles`, article bodies aur legal pages ka content plan banate waqt fetch nahi ho saka. A6 prompt unhe scrape karega.
11. **Template license:** Webangles ek Framer template hai. Layout ko apne code mein recreate karne se pehle uska license/purchase terms check karein.

4. Roadmap

Phase	Kaam	Prompts	Done when
0 Setup	`agy`, repo, AGENTS.md, MCP, skills, hooks	A1, A3, A4	`/mcp` mein servers connected, hooks active
1 Design extract	Webangles ke tokens, sections, components	A2	`design-ref/` ki files ban jayein
2 Contract	Block schemas + content fit	C1, P1	`pnpm -r typecheck` pass
3 Database	Schema + migrations + seed	P2	Clean DB par migrations aur seed chalein
4 API	Auth, content, media, pages	P3, P4, P5, P6	vitest green, `docs/api.md` export
5 Content scrape	Subpages + assets	A6	`content/seed/` poora, assets downloaded
6 Admin + Web	Panel aur public site (parallel)	A5 (P7, P8)	Dono build hon, Playwright smoke pass
7 Forms + migration	Contact/newsletter, seed import	P9, P10	Seeded site poora render ho
8 Visual QA	Template se fidelity check	P13	Diff report mein koi major gap na ho
9 Deploy + security	Server config, audit	P11, P12	Critical items fix
10 Launch	Checklist section 15	—	DNS cutover aur monitoring

Har phase ke liye alag git branch aur alag `agy` session rakhein.

5. Antigravity CLI workflow

Setup

```
curl -fsSL https://antigravity.google/cli/install.sh | bash
agy --version
git init && git checkout -b chore/setup
```

Pehli baar `agy` chalane par browser mein Google login (OAuth) khulta hai.

Project files

- `AGENTS.md`: agent ke rules (purani `GEMINI.md` bhi chalti hai, lekin `AGENTS.md` recommended hai).
- `.agents/skills/`: repeatable workflows (A3).
- `.agents/mcp_config.json`: project-level MCP servers. Playwright MCP (browser, screenshots, DOM extraction, UI tests) aur Postgres MCP (local DB dekhne ke liye) add karein, phir `/mcp` se verify karein.
- Global config ka folder alag sources mein alag likha hai (`~/.gemini/antigravity-cli/` aur `~/.antigravity/`). Apni machine par `/config` se confirm karein.

Har phase ka loop

```
agy                        # naya session, ek phase = ek session
/grill-me                  # bade module se pehle sawal-jawab
/plan <prompt>             # plan banwayein, /artifacts mein review karein
(approve)
/goal <done condition>     # jab tak tests/build pass na ho
/diff                      # review
! git add -A && git commit -m 'feat(...): ...'
```

Lambe tasks (build, test) ke liye `Ctrl+b` aur `/tasks`. Context bharne lage to `/context` check karein aur naya session shuru karein. Session ki yaad `PROGRESS.md` se chalegi.

Permission modes

Kaam	Mode
Scaffold, design extract, frontend	`proceed-in-sandbox`
Auth, DB, API, migrations	`request-review` (default)
Deploy scripts, security fixes	`strict`

`--dangerously-skip-permissions` is repo par kabhi use na karein.

Models: routine kaam (components, CRUD) ke liye Gemini Flash. Architecture, auth, DB schema (P2, P3, P6) aur security review (P12) ke liye Pro ya Claude model, jo bhi `/model` mein available ho.

Quota: ek July 2026 ke article ke mutabiq free tier roz ke ~20 agent requests tak limited hai. `/goal` loops bhi requests kharch karte hain. Apne account ka plan check karein, aur chhote chhote prompts ki jagah bade, achhe specified prompts chalayein.

Safety: `agy` ko production server ya prod credentials ke saath na chalayein. `.env*` files agent se door rakhein. P11 mein agent sirf deploy files likhe, server par aap khud chalayein.

6. Repo aur environment variables

api: `DATABASE_URL`, `COOKIE_SECRET`, `ACCESS_TTL`, `REFRESH_TTL`, `CORS_ORIGINS`, `UPLOADS_DIR`, `MEDIA_BASE_URL`, `WEB_REVALIDATE_URL`, `REVALIDATE_SECRET`, `PREVIEW_SECRET`, `SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASS`, `MAIL_TO`, `TURNSTILE_SECRET` (optional), `ADMIN_SEED_EMAIL`, `ADMIN_SEED_PASSWORD`.

web: `NEXT_PUBLIC_SITE_URL`, `API_BASE_URL`, `REVALIDATE_SECRET`, `PREVIEW_SECRET`.

admin: `VITE_API_BASE_URL`.

Har app mein `.env.example` ho aur startup par zod se env validate ho.

7. Data model

Table	Key columns
users	id, email, password_hash, role (admin/editor), is_active, last_login_at
sessions	id, user_id, refresh_hash, expires_at, user_agent, ip
media	id, filename, mime, size, width, height, blurhash, alt, folder, variants jsonb, uploaded_by
site_settings (singleton)	site_name, tagline, logo_id, favicon_id, seo jsonb, contact jsonb, socials jsonb, footer jsonb
navigation (singleton)	items jsonb (label, href, children[]), cta jsonb
pages	id, slug, title, seo jsonb, status, published_at
page_sections	id, page_id, type, position, enabled, data jsonb
services	id, number, title, slug, description, bullets jsonb, cta jsonb, position, status
projects	id, title, slug, category, summary, year, client, tags jsonb, cover_id, gallery jsonb, url, body jsonb, featured, position, status, seo jsonb

Table	Key columns
testimonials	id, name, role, company, quote, rating, avatar_id, position, status
posts	id, title, slug, category, excerpt, body jsonb, cover_id, author, read_minutes, published_at, status, seo jsonb
team_members	id, name, role, photo_id, bio, socials jsonb, position, status
clients	id, name, logo_id, href, position
tech_stack_groups	id, title, items jsonb, position
faqs	id, question, answer, position, status
form_submissions	id, type, payload jsonb, status (new/read/archived), ip, created_at
subscribers	id, email, confirmed, created_at
audit_log	id, user_id, action, entity, entity_id, diff jsonb, created_at
revisions	id, entity, entity_id, data jsonb, created_by, created_at

Sab tables mein UUID primary key, `created_at`, `updated_at`. Content tables mein `status` (draft/published) aur unique `slug`. Rich text Tiptap JSON mein store hoga.

8. Section blocks (contract)

Har block ka zod schema `packages/shared` mein hoga. Fields ki final lengths A2 ke slot character counts se C1 prompt tay karega.

- **hero:** eyebrow, headline, description, primaryCta {label, href}, secondaryCta {label, href}, scrollLabel, heroMedia (optional).
- **clientMarquee:** label (optional), items[] {name, logo (optional), href (optional)}, speed, direction.
- **aboutStats:** eyebrow, heading, paragraphs[], points[] (3), stats[] {value, suffix, label}, badges[] {value, label}, cta (optional), image (optional).
- **services:** eyebrow, heading, intro, serviceIds[] (services collection se), marqueeTags[], layout.
- **whyUs:** heading, items[3] {title, description (optional), icon}.

- **techStack:** eyebrow, heading, intro, groupIds[] (tech_stack_groups se), showMarquee.
- **featuredProjects:** eyebrow, heading, filters[], limit, cta (optional), projectIds[] (optional manual pick).
- **testimonials:** heading, subheading, testimonialIds[], showRating.
- **journal:** eyebrow, heading, limit, allArticlesCta {label, href}.
- **cta:** eyebrow, heading, description, email, primaryCta, secondaryCta, badge.
- **contactForm:** heading, intro, details {phone, email, website, address}, fields[] (name, type, required), submitLabel, successMessage.

Har block mein common fields: `anchorId`, `enabled`, `theme` (light/dark variant, optional).

9. API surface

Public (read-only, published content, cache headers)

- `GET /api/public/site` (settings + navigation + footer)
- `GET /api/public/pages/:slug` (enabled sections, referenced collections resolved)
- `GET /api/public/projects?category=&limit=&cursor=` aur `GET /api/public/projects/:slug`
- `GET /api/public/posts?limit=&cursor=` aur `GET /api/public/posts/:slug`
- `GET /api/public/services`, `/testimonials`, `/team`, `/faqs`, `/clients`, `/tech-stack`
- `GET /api/public/sitemap`
- `POST /api/forms/contact`, `POST /api/forms/subscribe`

Admin (cookie auth + CSRF header + role check)

- `POST /api/auth/login`, `/logout`, `/refresh`, `GET /api/auth/me`
- `GET|POST /api/admin/:collection`, `GET|PUT|DELETE /api/admin/:collection/:id`
- `POST /api/admin/:collection/:id/publish`, `/unpublish`, `/duplicate`, `POST /api/admin/:collection/reorder`
- `GET|PUT /api/admin/singletons/:key` (settings, navigation, footer)
- `GET|PUT /api/admin/pages/:slug`, plus section add, remove, reorder, toggle
- `POST /api/admin/preview-token`
- `GET|POST|PATCH|DELETE /api/admin/media`
- `GET|PATCH|DELETE /api/admin/submissions`, `GET /api/admin/submissions/export.csv`
- `GET|POST|PATCH /api/admin/users`, `GET /api/admin/audit`

Error format: `{ error: { code, message, details } }`. List endpoints mein pagination, search aur sort.

10. Admin panel spec

- **Screens:** login, dashboard (recent submissions, drafts, quick links), generic collection list + editor, singleton editors (settings, navigation, footer), page section builder, media library, submissions inbox, users, audit log.
- **Roles:** `admin` sab kuch. `editor` content aur media, lekin users, settings aur deploy-related cheezen nahi.
- **Forms:** zod schema + `fieldMeta` se auto-generate. Widgets: text, textarea, Tiptap rich text, image picker, link, select, toggle, number, array (drag reorder), reference picker.
- **Section builder:** dnd-kit se reorder, enable/disable, block registry se add, har block ka form, live preview iframe (preview token ke saath).
- **UX rules:** unsaved-changes guard, draft aur publish buttons, slug auto-generate, publish confirm, revisions se restore, toasts, keyboard save (Ctrl+S), mobile par bhi usable.
- **Media picker:** upload, search, alt text required, folder filter, delete se pehle usage check.

11. Web frontend spec

Routes (purane URLs rakhein taake SEO ya redirects na toote): `/`, `/services`, `/portfolio`, `/portfolio/[slug]` (naya), `/team`, `/articles`, `/blog/[slug]`, `/careers`, `/contact`, `/privacy-policy`, `/terms-and-conditions`, `/cookie-policy`, 404. Home par section ids `about`, `skills`, `portfolio`, `contact` rakhein kyunke purane links `/#about` jaise hain.

Home ke sections aur Webangles ka reference

#	Block	Webangles reference	Note
1	hero	Hero	CodeXplus headline lambi hai (59 chars), wrap check karein
2	clientMarquee	Hero ke neeche tag marquee	Client names ke liye
3	aboutStats	About Us	Stats row naya add hoga
4	services	Services cards + tag marquee	6 cards se 4 (2x2), bullets marquee tags banenge

#	Block	Webangles reference	Note
5	whyUs	24/7 Support (3 points)	Teen differentiators
6	techStack	Features bento + Integrations logos	4 groups cards, items logo/chip row
7	featuredProjects	Portfolio	Filter chips naye hain
8	testimonials	Testimonials	6 items
9	journal	Koi equivalent nahi	Portfolio card style reuse
10	cta	Launch CTA / Final CTA	
11	contactForm	Contact page	Form naya
—	footer	Footer	

Hata dein: Pricing, Comparison, Process, About Nubien blocks, FAQ (optional page baad mein). Team `/team` page par jayegi.

Common requirements: ISR + cache tags (`revalidateTag` webhook se), preview mode token se, `next/image` + blurhash, per-page metadata aur OG, `sitemap.xml` aur `robots.txt` API se, `prefers-reduced-motion` respect, a11y (focus states, contrast, semantic headings), performance targets (Lighthouse 90+), breakpoints 390/768/1024/1440.

12. Prompts

Order: A1, A3, A4 (setup) → A2 (design extract) → C1 + P1 (contract) → P2 → P3–P6 (API) → A6 (scrape) → A5 (admin + web parallel, P7 + P8) → P9, P10 → P13 → P11, P12.

A1 — AGENTS.md (repo root par save karein)

```
# CodeXplus — Agent rules

## Project
Monorepo (pnpm): apps/api (Fastify 5, TypeScript strict), apps/admin (Vite +
React), apps/web (Next.js 15 App Router), packages/shared (zod schemas +
inferred types).
Goal: rebuild codexplus.site on the Webangles template layout, with a custom
Fastify backend and a custom admin panel that controls the whole site
(replacing Sanity). Full plan: docs/PLAN.md. Design reference: design-ref/.
Content: content/seed/.

## Conventions
```

```
- zod schemas in packages/shared are the single source of truth for
validation, admin forms, and types.
- No any. Env validated with zod at startup. Error shape: { error: { code,
message, details } }.
- Postgres via Drizzle with migrations; never edit an applied migration.
- Public API is read-only and returns published content only. Never expose
password hashes or internal fields.
- Conventional commits. Ask before adding dependencies outside the approved
list in docs/PLAN.md section 2.

## Operating rules for the agent
- One phase per session. Before finishing: run typecheck, lint, tests; update
PROGRESS.md (done / next / open decisions); commit on the phase branch.
- Never read, print, or edit .env* files or commit secrets; use .env.example.
- Never run destructive DB commands (drop/truncate/reset) except on the local
docker test DB.
- For any change to packages/shared, list affected apps and update them in
the same task.
- After the API phase, export OpenAPI and write docs/api.md as the contract
for admin and web.
- Never copy text from the Webangles template; only layout, spacing,
typography and motion are references.
- When unsure about a design decision, write it under Open decisions in
PROGRESS.md and ask instead of guessing.
```

A2 — Design extraction (Playwright MCP)

```
Using the Playwright MCP, open https://webangles.framer.website/ at 1440,
1024, 768 and 390 px widths. For each width save a full-page screenshot to
design-ref/<width>.png. Then, via page.evaluate, extract for every top-level
section in DOM order: its text slots (tag + character count only) and
computed styles (font-family, size, weight, line-height, letter-spacing,
color, background, border-radius, padding, gap, max-width, box-shadow).
Scroll the page and record animations and marquee behavior from
transform/animation/transition styles. Write: design-ref/tokens.json
(palette, type scale, spacing, radii, shadows, container width), design-
ref/sections.md (ordered section inventory, layout description, slot
character counts), design-ref/components.md (navbar, buttons, cards, marquee,
tabs, footer, hover states). Do not copy template text beyond slot lengths.
Do not invent values you could not measure.
```

A3 — Skills

```
Create these project skills under .agents/skills/. First check Antigravity's skill format in its docs, then write them:
1. add-block-type: add a page section block end to end: zod schema + fieldMeta in packages/shared, registry entry, API validation test, admin form auto-render check, web component in apps/web/components/sections, seed example.
2. add-collection: new collection end to end: Drizzle table + migration, shared schema, CRUD via the module factory, public route, admin screen, seed.
3. db-migration: generate, review the SQL, apply on a clean DB, never edit applied migrations.
Each skill ends with a verification checklist (typecheck, vitest, build).
```

A4 — Hooks

```
Set up hooks (check /hooks and the docs for the format): after any edit in apps/* or packages/*, run typecheck for the affected workspace; before finishing a task, run lint + tests for the workspace touched. Block edits to .env* files and to already-applied migration files. Show me the hook config before enabling it.
```

A5 — Admin aur web parallel (jab schemas aur docs/api.md freeze hon)

```
packages/shared and the API contract (docs/api.md) are frozen. Run two parallel subagents: (1) apps/admin using this spec: [paste P7], (2) apps/web using this spec plus design-ref/: [paste P8]. Neither may modify packages/shared or apps/api; if a contract change is needed, write it to docs/contract-requests.md and stop that task. Each works on its own branch (feat/admin, feat/web) and updates PROGRESS.md. Report status via /agents.
```

A6 — Subpages aur assets scrape (Playwright MCP)

```
Using the Playwright MCP, scrape https://codexplus.site/ and these pages:
/portfolio, /services, /team, /careers, /articles, every /blog/<slug> linked
from /articles, /contact, /privacy-policy, /terms-and-conditions, /cookie-
policy. For each page save the rendered text structure (headings, paragraphs,
lists, links) as JSON under content/seed/pages/<slug>.json. Extract
collections into content/seed/projects.json, posts.json (full body as
structured blocks), team.json, services.json, testimonials.json,
clients.json, tech-stack.json with fields matching packages/shared schemas.
Download every image and asset (including those on cdn.sanity.io) into
content/seed/assets/ with a manifest content/seed/assets.json (original URL,
local file, alt text if present, where used). Do not invent anything: write
every missing or placeholder-looking value (lorem ipsum, example domains,
zero counters) to content/seed/REPORT.md with the page and field.
```

C1 — Content fitting

```
Template section structure: [paste design-ref/sections.md]
Source content: docs/PLAN.md section 13 (and content/seed/ if present).

For every block in docs/PLAN.md section 8, map the source content to the
template slots. Rules:
- Keep the CodeXplus meaning and voice; do not invent facts, numbers,
clients, or claims.
- Stay within +/-15% of the template slot character counts; where the source
is longer (for example the hero headline), propose a trimmed variant AND
recommend font-size or layout changes instead of silently cutting.
- Output a table: Block | Field | Final text | Char count | Slot limit.
- Flag anything missing, placeholder-looking, stale or inconsistent (emails,
phone, address, dates, counters) instead of filling it in.
```

P1 — Scaffold + shared schemas

```
Set up the pnpm monorepo per AGENTS.md: tsconfig base, eslint, prettier, workspace scripts (dev, build, typecheck, lint, test). In packages/shared define zod schemas + inferred types for: Media, Seo, Link, SiteSettings, Navigation, Footer, Service, Project, Testimonial, Post, TeamMember, Client, TechStackGroup, Faq, ContactSubmission, form payloads. Then define the Section block registry as a discriminated union on type for: hero, clientMarquee, aboutStats, services, whyUs, techStack, featuredProjects, testimonials, journal, cta, contactForm. Use docs/PLAN.md section 8 for fields and this slot-length table: [paste C1 output]. Export a fieldMeta helper so each field can carry admin UI hints (label, widget, help text, maxLength). Add unit tests that parse sample data from content/seed/.
```

P2 — Database

```
In apps/api create the Drizzle Postgres schema + initial migration for all tables in docs/PLAN.md section 7. Use UUID primary keys, created_at/updated_at, status enum (draft|published), published_at, unique slugs, integer position for ordering, and indexes on slug, status, position and (page_id, position). Add docker-compose.yml for local Postgres, a migrate script, and a seed script that creates the first admin from ADMIN_SEED_EMAIL / ADMIN_SEED_PASSWORD (argon2id). Tests: migrations apply on a clean DB and the seed is idempotent.
```

P3 — Fastify core + auth

```
Build the Fastify 5 app in apps/api with a plugin architecture: config (zod env), db, zod type provider, pino logging, @fastify/helmet, cors (allowlist from env), rate-limit, cookie, swagger (dev only), central error handler using the standard error shape. Auth module: login, logout, refresh with rotation and reuse detection, me. argon2id hashing, httpOnly + Secure + SameSite=Lax cookies, strict rate limit on login, roles (admin, editor) enforced by a preHandler. Add a CSRF guard (custom header required on state-changing admin requests). Write vitest tests against a test DB for login, refresh rotation, role checks and rate limiting.
```

P4 — Content modules

```
Create a reusable CRUD module factory in apps/api for the collections
(services, projects, testimonials, posts, team_members, clients,
tech_stack_groups, faqs). Admin routes under /api/admin/*: list (pagination,
search, sort, status filter), get, create, update, delete, publish/unpublish,
reorder (bulk positions), duplicate. Public routes under /api/public/*:
published only, cache headers, filters. Validate with the shared zod schemas,
auto-generate slugs, write audit_log on every mutation, store a revision on
every update. Add singleton endpoints for site_settings, navigation, footer.
Tests for each module's happy path and authz.
```

P5 — Media

```
Add a media module: multipart upload with size limits, detect the real mime
via magic bytes (allow jpeg/png/webp/avif/sanitized svg/pdf), process with
sharp into webp + avif variants (400/800/1600) + blurhash + dimensions, store
files under UPLOADS_DIR with hashed names, DB row with alt text, folder and
uploader. Endpoints: upload, list/search, update alt/folder, delete (block if
referenced by any content). Return absolute public URLs via MEDIA_BASE_URL.
Tests for mime spoofing, oversize files and delete-when-referenced.
```

P6 — Pages, sections, revalidation, preview

```
Add pages + sections endpoints: get/update a page with ordered sections; add,
remove, reorder and enable/disable a section; each section's data validated
against the block registry in packages/shared. The public endpoint returns a
page with enabled sections only, with referenced collections resolved
(services, projects, testimonials, posts, tech stack). On publish, POST a
signed (HMAC) webhook to WEB_REVALIDATE_URL with cache tags. Add short-lived
signed preview tokens so the web app can render drafts. When done, export
OpenAPI and write docs/api.md with example requests and responses for every
endpoint.
```

P7 — Admin panel

```
Build apps/admin with Vite + React + TS, shadcn/ui, TanStack Router + Query, react-hook-form + zod resolver. Screens: login, dashboard, generic collection list + editor, singleton editors (settings, navigation, footer), media library with picker, page section builder, submissions inbox, users, audit log. Forms must be generated from the shared zod schemas + fieldMeta (text, textarea, Tiptap rich text, image picker, link, select, toggle, number, array with drag reorder, reference picker). Section builder: dnd-kit reorder, enable/disable, add from the block registry, live preview iframe using the preview token. Add unsaved-changes guard, draft/publish buttons, revisions restore, toasts, Ctrl+S, role-based UI, and responsive layout. Use docs/api.md as the contract.
```

P8 — Web frontend (Webangles layout)

```
Build apps/web (Next.js 15 App Router, Tailwind, Framer Motion) recreating the layout and visual style of the Webangles template using design-ref/tokens.json, sections.md and components.md as the spec (do not copy template text). Render pages by mapping API section types to components (Hero, ClientMarquee, AboutStats, Services, WhyUs, TechStack, FeaturedProjects, Testimonials, Journal, Cta, ContactForm), all data from the public API, nothing hardcoded. Routes per docs/PLAN.md section 11. ISR with cache tags and a /api/revalidate route that verifies the HMAC secret. Preview mode via token. next/image with the media variants + blurhash, metadata from API SEO fields, sitemap.xml and robots.txt from the API. Respect prefers-reduced-motion. Test at 390/768/1024/1440. Use docs/api.md as the contract.
```

P9 — Forms + email

```
Add POST /api/forms/contact and /api/forms/subscribe: zod validation, honeypot, per-IP rate limit, optional Cloudflare Turnstile, store in DB, send a notification email via SMTP (env), return a generic success. Admin inbox: new/read/archived, detail view, CSV export, delete. Wire the web ContactForm and newsletter form to these endpoints with accessible error and success states.
```

P10 — Seed import

```
Write a seed/import script that loads content/seed/ into the new DB: register every file in content/seed/assets/ through the media pipeline (variants, blurhash, alt), then create site settings, navigation, footer, the home page with sections in the order from docs/PLAN.md section 11, and all collections with media references resolved. Idempotent (safe to re-run). Do not invent missing data; print a report of every field that is still empty or marked TODO (see docs/PLAN.md section 14).
```

P11 — Deploy (Ubuntu)

```
Create deploy config for Ubuntu (files only, do not run anything on a server): Nginx server blocks for www, admin and api (TLS via certbot, gzip/brotli, caching for /uploads, security headers, client_max_body_size), a PM2 ecosystem file for api and web, admin built as static files, Postgres setup notes, a nightly pg_dump + uploads backup with retention, a GitHub Actions workflow that deploys over SSH with zero-downtime reload and a migration step, and a rollback procedure in docs/deploy.md.
```

P12 — Security + QA review

```
Review the whole repo as a security auditor and QA lead: authz on every admin route, mass-assignment, upload abuse, XSS in rich text rendering, cookie/CORS config, CSRF guard, rate limits, secrets handling, SQL injection surface, dependency audit, and test coverage gaps. Output a prioritized list with exact file/line fixes, then fix the critical ones.
```

P13 — Visual QA against the template

```
Using the Playwright MCP, run the local web app. For widths 1440, 1024, 768 and 390, screenshot each home section and compare it with the matching section in design-ref/<width>.png (layout, spacing, type scale, colors, alignment, hover states). Ignore text differences; compare layout and styling only. Output a table: Section | Width | Deviation | Severity | Fix. Fix all High items, re-run, and save before/after screenshot paths in docs/visual-qa.md.
```

13. Content (seed) — codexplus.site ka poora homepage content

Yeh content codexplus.site ke homepage se liya gaya hai aur block fields (section 8) ke hisaab se rakha gaya hai. `TODO` ka matlab hai ke source mein asli value nahi mili ya placeholder thi. Slot lengths A2 ke baad C1 prompt se final honge, isliye yeh text v1 hai.

13.1 Site settings

- **siteName:** CodeXplus
- **seo.title:** CodeXplus — Digital Creative & Engineering Powerhouse
- **seo.description:** CodeXplus is a creative studio crafting brand identities, websites and full-stack digital products that move the world. Design. Engineering. Imagination.
- **logo / ogImage:** abhi cdn.sanity.io par hain. A6 inhe download karega.
- **contact.email:** TODO (hello@codexplus.studio ya info@codexplus.site)
- **contact.phone:** +92-3243315524 (WhatsApp: https://wa.me/+923243315524)
- **contact.address:** R-591, F.B Area Block 20, Karachi, Pakistan
- **socials:** GitHub, LinkedIn, Twitter, Instagram, Dribbble — TODO asli profile URLs (abhi generic hain)
- **credit:** Develop by ARMAN NAQVI (https://techmiresolutions.com/)

13.2 Navigation

Maujooda menu: About, Projects, Contact. Menu panel ke groups:

- **About:** Company → `/#about`, Services → `/services`, Careers → `/careers`
- **Projects:** Featured Work → `/portfolio`, Tech Stack → `/#skills`, Case Studies → `/portfolio`
- **Contact:** Get In Touch → `/contact`, Email → mailto (TODO final email), LinkedIn, GitHub

Template ke style mein flat nav ka suggestion (aapka faisla): Home, About, Services, Projects, Journal, Contact + CTA button `Get In Touch`.

13.3 hero

- **eyebrow:** Digital Creative & Engineering Powerhouse (site title se liya, confirm karein)
- **headline:** Transforming Bold Ideas Into Powerful Digital Products.
- **description:** CodeXPlus is an independent creative studio. We combine imagination, full-stack craft, and digital design expertise to build products, brands, and websites that people remember.
- **primaryCta:** View Selected Work → `#portfolio`
- **secondaryCta:** Get In Touch → `#contact`
- **scrollLabel:** Scroll Down → `#about`

13.4 clientMarquee

- **items:** NORDIC MOUNTS, LUMEN LABS, FERN & CO, VERTEX PAY, ORBIT TRAVEL, HALCYON, MARROW, PAPER PLANE (logos TODO, abhi sirf text)

13.5 aboutStats

- **eyebrow:** ABOUT THE STUDIO
- **heading:** Where imagination meets engineering.
- **paragraphs:**
  1. At our design studio, we're passionate about transforming raw ideas into reality. Every project starts with curiosity and ends with something people genuinely love to use.
  2. From first sketch to final deploy, we grow together with our clients — pairing clean, thoughtful engineering with visual direction that refuses to blend in.
- **points:** Collaborative, senior-led delivery / Clean, maintainable engineering / Uncompromising attention to detail
- **badges:** 12+ YEARS OF CRAFT / 40k+ creators shipped (confirm karein ke yeh claim sahi hai)
- **stats:** Projects Completed: TODO (+) / Client Retention: TODO (%) / Global Awards: TODO / Years of Craft: 12+

13.6 services

- **eyebrow:** What we do
- **heading:** Services engineered for impact.
- **intro:** Four tightly-argued disciplines, one accountable team. Strategy through shipping — no hand-offs, no dropped balls.
- **marqueeTags:** har service ke bullets (neeche)

01 — Branding & Identity Design

- description: End-to-end visual identities engineered to be unforgettable — logo systems, typography, art direction and brand guidelines that scale from business card to billboard.
- bullets: Brand strategy & positioning / Logo systems & guidelines / Art direction / Design systems
- cta: Start a project → `/#contact`

02 — Website Design & Development

- description: Blazing-fast, conversion-focused websites built on Next.js and modern full-stack architecture — with headless CMS, animations and pixel-perfect responsive craft.
- bullets: Next.js & full-stack builds / Headless CMS integration / 3D WebGL & interactive motion / SEO & Core Web Vitals
- cta: Start a project → `/#contact`

03 — Advertising & Marketing Campaigns

- description: Performance-driven campaigns that turn attention into revenue — from SEO foundations and content strategy to paid media and continuous optimization.
- bullets: SEO & performance audits / Campaign creative & art direction / Analytics, CRO & Funnels / Omnichannel Growth
- cta: Start a project → `/#contact`

04 — Creative Consulting & Development

- description: Senior-level thinking on tap. We audit, mentor and co-build alongside your team — turning ambiguous briefs into shipped, measurable product outcomes.
- bullets: Product & UX consulting / Tech architecture & mentorship / Design sprints & prototyping / Scalable cloud deployment
- cta: Start a project → `/#contact`

13.7 whyUs

- **heading:** TODO (source mein alag heading nahi hai; About ke 3 points reuse ho rahe hain)
- **items:** Collaborative, senior-led delivery / Clean, maintainable engineering / Uncompromising attention to detail (descriptions optional, TODO)

13.8 techStack

- **eyebrow:** Skills & Stack
- **heading:** Engineered with precision & modern technology.
- **intro:** A curated tech stack focused on high performance, seamless user experience, and scalable code standards.
- **groups (maujooda text; section 3 point 5 dekhein):**
  - Frontend Engineering: React.js, Next.js 15, TypeScript, Tailwind CSS, Three.js / WebGL, Framer Motion, GSAP
  - Backend & CMS: Node.js, Sanity CMS, REST / GraphQL, PostgreSQL, Next API Routes, Server Components

- UI/UX & Product Design: Figma, Design Systems, Wireframing, Interactive Prototypes, Micro-interactions, User Research
- Performance & Workflow: Vercel, Git / GitHub, SEO Optimization, Web Vitals, CI/CD, Responsive Architecture

13.9 featuredProjects

- **eyebrow:** Selected work
- **heading:** Featured projects.
- **filters:** All, Web Design, E-Commerce, Product Design, Brand Identity, Web App, Mobile App
- **items:** TODO, projects ka data `/portfolio` se A6 scrape karega.

13.10 testimonials

- **heading:** Client Testimonial
- **subheading:** Trusted by ambitious founders, teams, and industry leaders worldwide.
- **rating:** 4.9 (sab ke liye)

Name	Role	Company	Quote
Elena Vasquez	Founder	Maison Noor	CodeX+ didn't just design a website — they rebuilt how our entire company thinks about its brand. Every pixel feels deliberate. It changed the business.
Marcus Feld	CPO	Vanta Commerce	The fastest, most considered build we've ever run. Conversion up 38% in one quarter, and the dev experience is genuinely beautiful.
Sofia Lindgren	VP Product	Pulse Analytics	They translate vision into craft better than any team we've worked with. Our dashboard finally feels premium enough for our clients.
Jonas Weber	CEO	Northwind	From strategy to shipping, a single accountable partner. Our rebrand doubled wholesale inquiries in six months. Unreal.

Name	Role	Company	Quote
Aisha Bello	Head of Design	Folio	We shipped a full product from zero in four months. The care they put into motion and micro-interaction is what makes it feel expensive.
Tomás Rivera	CEO	Aurora Health	Design that genuinely calms people down. Our retention numbers prove it. CodeX+ sets the bar for health-tech craft.

13.11 journal

- **eyebrow:** Journal
- **heading:** Insights & field notes.
- **allArticlesCta:** All articles → `/articles`

Category	Date	Read	Title	Excerpt	Slug
Design	Jun 18, 2026	8 min	Why Design Systems Fail After Year One	Most design systems die quietly. Here's the organizational wiring — not the tokens — that keeps them alive.	why-design-systems-fail-after-year-one
Engineering	May 02, 2026	12 min	The Next.js Performance Checklist We Ship To Every Client	Image pipelines, edge caching, font strategy and the 190ms storefront recipe we apply before launch day.	the-next-js-performance-checklist-we-ship-to-every-client
Motion	Mar 21, 2026	6 min	Micro-Motion: The 12 Principles of Interfaces That Feel Alive	A tiny easing curve is worth a thousand hero videos. Notes from our design systems on making UI breathe.	micro-motion-the-12-principles-of-interfaces-that-feel-alive

Article bodies A6 scrape karega.

13.12 cta

- **eyebrow:** Let's build something
- **heading:** Have an idea worth building?
- **description:** Tell us where you want to go. We'll bring the strategy, the craft and the code. Response within 24 hours.

- **email:** info@codexplus.site (TODO final email)
- **primaryCta / secondaryCta:** See the work → `#portfolio`
- **badge:** Currently booking Q3 2026 projects — TODO stale hai, naya quarter likhein.

13.13 contactForm

- **heading:** Contact Us
- **intro:** TODO (maujooda text lorem ipsum hai)
- **details:** phone, email, website, address — TODO (maujooda values template placeholders hain, section 3 point 2)
- **fields:** Name, Email, Phone Number, Message (textarea)
- **submitLabel:** SUBMIT

13.14 footer

- **newsletterLabel:** Subscribe our newsletter:
- **links:** Home `/`, Portfolio `/portfolio`, Services `/services`, Team `/team`, Contact `/contact`, Blog `/articles`, About `/#about`
- **offices:** Pakistan — R-591, F.B Area Block 20, Karachi, +92-3243315524 (WhatsApp). Worldwide Delivery — Digital Operations & Cloud, Global Client Hub, +92-3243315524.
- **legal:** Privacy Policy `/privacy-policy`, Terms and conditions `/terms-and-conditions`, Cookie Policy `/cookie-policy`, Careers `/careers`
- **copyright:** © Copyright 2026 - CodeXplus. All Rights Reserved. Develop by ARMAN NAQVI.

13.15 Jo content abhi nahi mila

Projects (titles, categories, images, descriptions), Team members, Careers page, Services page ki detail, article bodies, legal pages ka text, aur FAQs (site par FAQ nahi hai). Yeh sab A6 se aayega.

14. Content TODO list

#	Item	Kahan	Kis se chahiye
1	Projects Completed, Client Retention, Global Awards ke numbers	aboutStats.stats	Aap / client
2	Final email (hello@codexplus.studio ya info@codexplus.site)	settings, cta, contact	Aap

#	Item	Kahan	Kis se chahiye
3	Contact intro text, phone, website, address (placeholders hatana)	contactForm	Aap
4	Booking badge ka naya quarter	cta.badge	Aap
5	Tech stack group ka text naye stack ke hisaab se (Sanity, Next API Routes, Vercel)	techStack	Aap
6	Asli social profile URLs	settings.socials	Aap
7	Client logos (abhi sirf names)	clientMarquee	Client assets
8	Testimonials, client names aur claims ki confirmation	testimonials	Aap / clients
9	whyUs ki heading aur descriptions	whyUs	Aap (ya C1 se draft)
10	Eyebrow text confirm (Digital Creative & Engineering Powerhouse)	hero	Aap
11	Projects, team, careers, article bodies, legal pages	content/seed/	A6 scrape
12	Sanity ke saare images download	content/seed/assets/	A6 scrape
13	Webangles template license check	—	Aap

15. QA, deploy aur launch checklist

Functional: sab routes load hon, section builder se reorder/enable/disable ka asar public site par aaye, publish par revalidate chale, preview draft dikhaye, contact aur newsletter forms DB mein save hon aur email aaye, media delete usage check kare.

Visual: P13 report mein koi High deviation na ho. 390/768/1024/1440 par manual check bhi karein.

Performance: Lighthouse 90+ (performance, accessibility, SEO, best practices) home aur ek article page par. Images next/image se, fonts self-hosted ya next/font se.

SEO: har page ka title, description aur OG, `sitemap.xml`, `robots.txt`, canonical URLs, structured data (Organization, Article), purane URLs ka status 200.

Security (P12): authz har admin route par, cookies Secure + httpOnly, CSRF header, rate limits, upload mime checks, rich text sanitize, dependency audit, `.env` secrets sirf server par.

Ops: nightly Postgres backup + uploads backup, restore ek baar test karein, logs (pino) rotate hon, health endpoint, uptime monitor, admin ka pehla password change.

Launch steps

1. Staging par (`staging.codexplus.site` ya similar) poora stack chalayein aur content freeze se pehle ek baar A6 dobara chalakar naya content sync karein.
2. DNS TTL pehle se kam kar dein.
3. Production par deploy (P11 ke files), migrations aur seed import chalayein.
4. Search Console mein sitemap submit karein aur purane URLs check karein.
5. Cutover ke baad 24 se 48 ghante logs aur forms monitor karein.
6. Sanity ko kam az kam do hafte read-only rakhein. Jab sab images aur content apni media library mein confirm ho jayein, tab hi band karein. Repo mein `cdn.sanity.io` ka koi reference baqi na ho (grep se check).

16. URL aur redirect map

Plan ke mutabiq saare maujooda URLs wahi rehte hain, isliye redirects ki zaroorat nahi:

Maujooda URL	Naya URL
`/` aur `/#about`, `/#skills`, `/#portfolio`, `/#contact`	Same (home par section ids rakhein)
`/services`, `/portfolio`, `/team`, `/careers`, `/contact`	Same
`/articles`, `/blog/<slug>`	Same
`/privacy-policy`, `/terms-and-conditions`, `/cookie-policy`	Same
—	`/portfolio/<slug>` (naya project detail page)

Agar koi slug ya route badle to Nginx ya Next config mein 301 redirect lagayein aur is table ko update karein.

17. Sources

- https://codexplus.site/ (homepage content, nav, footer, metadata)
- https://webangles.framer.website/ (layout reference, section order)
- https://antigravity.google/docs/getting-started?tab=cli (official docs entry)
- https://docs.flutter.dev/ai/antigravity-cli (AGENTS.md, `.agents/skills/`, config notes)
- https://www.gradually.ai/antigravity-cli-befehle/ (commands, permission modes, MCP config paths, quota; July 2026)

Antigravity CLI ki details tezi se badal rahi hain, isliye `agy` ke `/help` aur official docs se confirm karte rahein.